

- partie1 du projet :

❑ **Config SW1:**

Building configuration... Current
configuration : 3041 bytes

!

version 12.4 service timestamps debug
datetime msec service timestamps log
datetime msec no service password-
encryption no service dhcp

!

hostname SW1

!

boot-start-marker boot-end-marker

!

!

no aaa new-model memory-size
iomem 5 no ip routing no ip
icmp rate-limit unreachable
no ip cef

!

!

!

!

no ip domain lookup

!

multilink bundle-name authenticated

!

!

!

!

!

!

!

!

!

!

!

!

```
!  
!  
!  
!  
!  
!  
!  
macro name add_vlan  
end vlan database  
vlan $v exit  
@ macro name  
del_vlan end vlan  
database no vlan $v  
exit  
@  
!  
vtp file nvram:vlan.dat  
archive log config  
hidekeys  
!  
!  
!  
!  
ip tcp synwait-time 5  
!  
!  
!  
!  
interface FastEthernet0/0 description *** Unused  
for Layer2 EtherSwitch ***  
no ip address no  
ip route-cache  
shutdown  
duplex auto  
speed auto  
!  
interface FastEthernet0/1 description *** Unused  
for Layer2 EtherSwitch *** no ip address no ip  
route-cache shutdown duplex auto speed auto  
!
```

```
interface FastEthernet1/0 switchport trunk
allowed vlan 1,2,10,20,1002-1005 switchport
mode trunk duplex full speed 100
!
interface FastEthernet1/1
switchport mode trunk
duplex full speed 100
!
interface FastEthernet1/2
duplex full speed 100
!
interface FastEthernet1/3
duplex full speed 100
!
interface FastEthernet1/4
duplex full speed 100
!
interface FastEthernet1/5
duplex full speed 100
!
interface FastEthernet1/6
duplex full speed 100
!
interface FastEthernet1/7
duplex full speed 100
!
interface FastEthernet1/8
duplex full speed 100
!
interface FastEthernet1/9
duplex full speed 100
!
interface FastEthernet1/10
duplex full speed 100
!
interface FastEthernet1/11
duplex full speed 100
!
interface FastEthernet1/12
duplex full speed 100
!
```

```

interface FastEthernet1/13
switchport access vlan 99
duplex full speed 100
!
interface FastEthernet1/14
switchport access vlan 10
duplex full speed 100
!
interface FastEthernet1/15
switchport access vlan 20
duplex full speed 100
!
interface Vlan1
no ip address no
ip route-cache
shutdown
!
ip forward-protocol nd
!
!
no ip http server no ip
http secure-server
! no cdp log mismatch duplex

```

The screenshot shows a terminal window with a network configuration. At the top, there are tabs for 'ESW0', 'ESW1', 'R6', and 'ESW5'. The main window displays a configuration for a switch module (likely a Catalyst 3560-E) connected to a router (R6). The configuration includes setting the switchport access VLANs (99, 10, 20) and duplex/full speed for interfaces FastEthernet1/13, 1/14, and 1/15. It also shows the configuration for the Vlan1 interface, including disabling IP, enabling route-cache, and shutting down the interface. The configuration is saved to the startup configuration. The terminal output shows the configuration being applied to the switch module.

```

R6#configure terminal
R6(config)#module slot1
R6(config-slot)#switchport access vlan 99
R6(config-slot)#interface FastEthernet1/13
R6(config-slot-if)#switchport access vlan 99
R6(config-slot-if)#duplex full
R6(config-slot-if)#speed 100
R6(config-slot-if)#exit
R6(config-slot)#interface FastEthernet1/14
R6(config-slot-if)#switchport access vlan 10
R6(config-slot-if)#duplex full
R6(config-slot-if)#speed 100
R6(config-slot-if)#exit
R6(config-slot)#interface FastEthernet1/15
R6(config-slot-if)#switchport access vlan 20
R6(config-slot-if)#duplex full
R6(config-slot-if)#speed 100
R6(config-slot-if)#exit
R6(config-slot)#interface Vlan1
R6(config-slot-if)#no ip address
R6(config-slot-if)#ip route-cache
R6(config-slot-if)#shutdown
R6(config-slot-if)#exit
R6(config-slot)#end
R6#copy running-config startup-config

```

```
ESW1#enable
ESW1#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
ESW1(config)#interface ?
Async                Async Interface
BVI                  Bridge-Group Virtual Interface
CDMA-Ix              CDMA Ix Interface
CTunnel              CTunnel Interface
Dialer                Dialer Interface
FastEthernet          FastEthernet IEEE 802.3
Group-Async           Async Group Interface
Lan                  Lan Interface
Loopback              Loopback Interface
MFR                  Multilink Frame Relay bundle Interface
Multilink             Multilink-group Interface
Null                 Null Interface
Port-channel          Ethernet Channel of Interfaces
Serial                Serial
Tunnel                Tunnel Interface
Vif                  PGM Multicast Host Interface
Virtual               Virtual Interface
Virtual-ppp           Virtual PPP Interface
Virtual-template       Virtual Template Interface
Virtual-tunneling      Virtual Tunneling
Vlan                  Vlan IEEE 802.1q
XtagATM               Extended Tag ATM Interface
range                 Interface range command
vml                   Virtual Multipoint Interface

ESW1(config)#interface fastEthernet 1/24
ESW1(config-if)#switchport mode access
ESW1(config-if)#switchport access vlan 20
ESW1(config-if)#interface fastEthernet 1/25
ESW1(config-if)#switchport mode access vlan 20
ESW1(config-if)#interface fastEthernet 1/26
ESW1(config-if)#switchport mode trunk
ESW1(config-if)#
*Mar 1 08:23:33.825: EDPF-S-TRUNKPORTON: Port fastEthernet 1/26 has become dot1q trunk
ESW1(config-if)#interface fastEthernet 1/27
ESW1(config-if)#switchport mode trunk
ESW1(config-if)#
*Mar 1 08:25:27.179: EDPF-S-TRUNKPORTON: Port fastEthernet 1/27 has become dot1q trunk
ESW1(config-if)#
```

```

!
!
!
!
!
control-plane
!
!
!
!
!
!
!
!
!
!
banner exec ^C
*****

This is a normal Router with a Switch module inside (NM-16ESW)
It has been pre-configured with hard-coded speed and duplex
To create vlans use the command "vlan database" in exec mode
After creating all desired vlans use "exit" to apply the config
To view existing vlans use the command "show vlan-switch brief"
Alias(exec) : vl - "show vlan-switch brief" command
Alias(configure): va X - macro to add vlan X
Alias(configure): vd X - macro to delete vlan X
*****

^C alias configure va macro global trace
add_vlan $v alias configure vd macro global
trace del_vlan $v alias exec vl show vlan-switch
brief
!
line con 0 exec-timeout
0 0 privilege level 15
logging synchronous
line aux 0 exec-timeout
0 0 privilege level 15
logging synchronous
line vty 0 4 login
!
!
end

```

!

VLAN Name Status Ports

----- 1

default active Fa1/2, Fa1/3, Fa1/4, Fa1/5

Fa1/6, Fa1/7, Fa1/8, Fa1/9

Fa1/10, Fa1/11, Fa1/12

10 students active Fa1/14

20 profs active Fa1/15

1002 fddi-default act/unsup

1003 token-ring-default act/unsup

1004 fddinet-default act/unsup 1005 trnet-default act/unsup **Config SW2:**

Building configuration... Current

configuration : 3008 bytes

!

version 12.4 service timestamps debug

datetime msec service timestamps log

datetime msec no service password-

encryption no service dhcp

!

hostname SW2

!

boot-start-marker boot-end-marker

!

!

no aaa new-model memory-size

iomem 5 no ip routing no ip

icmp rate-limit unreachable

no ip cef

!

!

!

no ip domain lookup

!

multilink bundle-name authenticated

!

!

!

!

!

!

!

```

!
!
!
!
!
!
!
!
!
!
!
!
macro name add_vlan
end vlan database
vlan $v exit
@ macro name
del_vlan end vlan
database no vlan $v
exit
@
!
vtp file nvram:vlan.dat
archive log config
hidekeys
!
!
!
.

```

Pour interconnecter les PC appartenant au même VLAN, vous devez configurer les ports des switchs comme des ports d'accès et leur attribuer le numéro du VLAN correspondant. Par exemple, pour le PC11, vous devez configurer le port du switch S1 comme un port d'accès au VLAN10. Voici un exemple de configuration pour le switch S1 :

```

S1(config)#interface fastEthernet 0/1
S1(config-if)#switchport mode access
S1(config-if)#switchport access vlan 10
S1(config-if)#exit

```


!

Pour interconnecter les PC appartenant à des VLAN différents, vous devez configurer les ports des switches comme des ports trunk et utiliser le protocole VTP pour propager les informations sur les VLAN. Par exemple, pour le serveur, vous devez configurer le port du switch S4 comme un port trunk et activer le VTP sur tous les switches. Voici un exemple de configuration pour le switch S4 :

```
S4(config)#vtp mode server
S4(config)#vtp domain projet
S4(config)#vtp password secret
S4(config)#interface fastEthernet 0/1
S4(config-if)#switchport mode trunk
S4(config-if)#exit
```

Pour que les PC puissent communiquer entre eux au travers des VLAN, vous devez également configurer le routage inter-VLAN sur les routeurs. Pour cela, vous devez créer des interfaces virtuelles ou sous-interfaces sur les routeurs et leur attribuer des adresses IP dans les plages des VLAN. Par exemple, pour le routeur R1, vous devez créer deux sous-interfaces pour les VLAN10 et VLAN20. Voici un exemple de configuration pour le routeur R1 :

```
R1(config)#interface fastEthernet 0/0
R1(config-if)#no ip address
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#interface fastEthernet 0/0.10
R1(config-subif)#encapsulation dot1Q 10
R1(config-subif)#ip address 172.16.10.1 255.255.255.0
R1(config-subif)#exit
R1(config)#interface fastEthernet 0/0.20
R1(config-subif)#encapsulation dot1Q 20
R1(config-subif)#ip address 172.16.20.1 255.255.255.0
R1(config-subif)#exit
```

Si vous voulez en savoir plus sur les VLAN et le routage inter-VLAN

```
!  
ip tcp synwait-time 5  
!  
!  
!  
!  
interface Port-channel1 switchport  
mode trunk  
!  
interface FastEthernet0/0 description *** Unused  
for Layer2 EtherSwitch ***  
no ip address no  
ip route-cache  
shutdown  
duplex auto  
speed auto  
!  
interface FastEthernet0/1 description *** Unused  
for Layer2 EtherSwitch ***  
no ip address no  
ip route-cache  
shutdown  
duplex auto  
speed auto  
!  
interface FastEthernet1/0  
switchport mode trunk  
duplex full speed 100  
channel-group 1 mode on  
!  
interface FastEthernet1/1  
switchport mode trunk  
duplex full speed 100  
channel-group 1 mode on  
!  
interface FastEthernet1/2  
duplex full speed 100  
!  
interface FastEthernet1/3  
duplex full speed 100
```

```
!  
interface FastEthernet1/4  
duplex full speed 100  
!  
interface FastEthernet1/5  
duplex full speed 100  
!  
interface FastEthernet1/6  
duplex full speed 100  
!  
interface FastEthernet1/7  
duplex full speed 100  
!  
interface FastEthernet1/8  
duplex full speed 100  
!  
interface FastEthernet1/9  
duplex full speed 100  
!  
interface FastEthernet1/10  
duplex full speed 100  
!  
interface FastEthernet1/11  
duplex full speed 100  
!  
interface FastEthernet1/12  
duplex full speed 100  
!  
interface FastEthernet1/13  
duplex full speed 100  
!  
interface FastEthernet1/14  
duplex full speed 100  
interface FastEthernet1/15  
duplex full speed  
100  
!  
interface Vlan1  
no ip address no  
ip route-cache  
shutdown  
!
```

```
!  
ip forward-protocol nd  
!  
!  
no ip http server no ip  
http secure-server  
!  
no cdp log mismatch duplex  
!  
!  
!  
!  
!  
!  
control-plane  
!  
!  
!  
!  
!  
!  
!  
!  
!  
banner exec ^C  
*****
```

This is a normal Router with a Switch module inside (NM-16ESW)
It has been pre-configured with hard-coded speed and duplex
To create vlans use the command "vlan database" in exec mode
After creating all desired vlans use "exit" to apply the config
To view existing vlans use the command "show vlan-switch brief"

!

```

R8
R8
Enter configuration commands, one per line. End with CNTL/Z.
R8(config)#int fastethernet 1/0
R8(config-if)#ip address 192.168.1.200
% Incomplete command.

R8(config-if)#ip address 192.168.1.200 255.255.255.0
R8(config-if)#int ethernet ?
% Unrecognized command
R8(config-if)#int ethernet 0/ ?
% Unrecognized command
R8(config-if)#int ethernet 0/0
R8(config-if)#ip address 192.168.1.200
% Incomplete command.

R8(config-if)#ip address 192.168.1.200 255.255.255.0
R8(config-if)#int serial 0/0
R8(config-if)#int ethernt 0/0
^
% Invalid input detected at '^' marker.

R8(config)#int ethernet 0/0
R8(config-if)#ip address 192.168.1.201 255.255.255.0
R8(config-if)#int serial 0/0
^
% Invalid input detected at '^' marker.

R8(config)#int serial 0/0
R8(config-if)#ip address 192.168.1.202 255.255.255.0
R8(config-if)#exit
R8(config)#exit
R8
*Mar  1 00:43:01.443: %SYS-5-CONFIG-I: Configured from console by console
R8#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              192.168.1.201   YES manual administratively down down
Serial0/0                 192.168.1.202   YES manual administratively down down
Serial0/1                 unassigned      YES unset  administratively down down
FastEthernet1/0          192.168.1.200   YES manual administratively down down
R8#
solarwinds | Solar-PuTTY (free tool) © 2019-2023 SolarWinds W

```

Alias(exec) : vl - "show vlan-switch brief" command

Alias(configure): va X - macro to add vlan X

Alias(configure): vd X - macro to delete vlan X

^C alias configure va macro global trace

add_vlan \$v alias configure vd macro global

trace del_vlan \$v alias exec vl show vlan-switch

brief

!

line con 0 exec-

timeout 0 0

privilege level 15

logging synchronous

line aux 0 exec-

timeout 0 0

privilege level 15

logging synchronous

line vty 0 4 login

!

!

end

VLAN Name Status Ports

----- 1

default active Fa1/2, Fa1/3, Fa1/4, Fa1/5

Fa1/6, Fa1/7, Fa1/8, Fa1/9

Fa1/10, Fa1/11, Fa1/12, Fa1/13

Fa1/14, Fa1/15

10 students active

20 profs active

1002 fddi-default act/unsup

1003 token-ring-default act/unsup

1004 fddinet-default act/unsup 1005 trnet-default act/unsup

Config SW3:

Building configuration... Current

configuration : 3031 bytes

!

version 12.4

service timestamps debug datetime msec

service timestamps log datetime msec

no service password-encryption no

service dhcp

```
!  
hostname SW3  
!  
boot-start-marker boot-end-marker  
!  
!  
no aaa new-model memory-size  
iomem 5 no ip routing no ip  
icmp rate-limit unreachable  
no ip cef  
!  
!  
!  
!  
no ip domain lookup  
!  
multilink bundle-name authenticated  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
macro name add_vlan end  
vlan database  
vlan $v exit
```

```
@ macro name
del_vlan end vlan
database no vlan $v
exit
@
!
vtp file nvram:vlan.dat
archive log config
hidekeys
!
!
!
!
ip tcp synwait-time 5
!
!
!
!
interface Port-channel2 switchport
mode trunk
!
interface FastEthernet0/0 description *** Unused
for Layer2 EtherSwitch ***
no ip address no
ip route-cache
shutdown
duplex auto
speed auto
!
interface FastEthernet0/1 description *** Unused
for Layer2 EtherSwitch ***
no ip address no
ip route-cache
shutdown
duplex auto
speed auto
!
interface FastEthernet1/0
switchport mode trunk
duplex full speed 100
!
```



```
interface FastEthernet1/1
switchport mode trunk
duplex full speed 100
channel-group 2 mode on
!
interface FastEthernet1/2
switchport mode trunk
duplex full speed 100
channel-group 2 mode on
!
interface FastEthernet1/3
duplex full speed 100
!
interface FastEthernet1/4
duplex full speed 100
!
interface FastEthernet1/5
duplex full speed 100
!
interface FastEthernet1/6
duplex full speed 100
!
interface FastEthernet1/7
duplex full speed 100
!
interface FastEthernet1/8
duplex full speed 100
!
interface FastEthernet1/9
duplex full speed 100
!
interface FastEthernet1/10
duplex full speed 100
!
interface FastEthernet1/11
duplex full speed 100
!
interface FastEthernet1/12
duplex full speed 100
!
interface FastEthernet1/13
duplex full speed 100
```

```
!  
interface FastEthernet1/14  
duplex full speed 100  
!  
interface FastEthernet1/15  
duplex full speed 100  
!  
interface Vlan1  
no ip address no  
ip route-cache  
shutdown  
!  
ip forward-protocol nd  
!  
!  
no ip http server no ip  
http secure-server  
!  
no cdp log mismatch duplex  
!  
!  
!  
!  
!  
!  
control-plane  
!
```

```
ESW6 ESW3 ESW1 R6
R6#config terminal
^
% Invalid input detected at '^' marker.

R6#config terminal
Enter configuration commands, one per line. End with CNTL/Z.
R6(config)#interface fa0/0
R6(config-if)#no shutdown
R6(config-if)#
*Mar 1 00:02:15.467: %LINK-3-UPDOWN: Interface FastEthernet0/0, change
*Mar 1 00:02:16.467: %LINEPROTO-5-UPDOWN: Line protocol on Interface F
R6(config-if)#int fa0.10
^
% Invalid input detected at '^' marker.

R6(config)#int fa0 0.10
^
% Invalid input detected at '^' marker.

R6(config)#int fa0/0.10
R6(config-subif)#encapsulation dot1q 10
^
% Invalid input detected at '^' marker.

R6(config-subif)#encapsulation dot1q 10
^
% Invalid input detected at '^' marker.

R6(config-subif)#encapsulation dot1q 10
R6(config-subif)#ip address 192.168.0.10 255.255.255.0
^
% Invalid input detected at '^' marker.

R6(config-subif)#ip address 192.168.0.10 255.255.255.0
R6(config-subif)#int fa0/0.20
R6(config-subif)#encapsulation dot1q 20
R6(config-subif)#ip address 192.168.0.20 255.255.255.0
192.168.0.0 overlaps with FastEthernet0/0.10
R6(config-subif)#
```

!

!

!

!

!

!

!

!

banner exec ^C

This is a normal Router with a Switch module inside (NM-16ESW)

It has been pre-configured with hard-coded speed and duplex

To create vlans use the command "vlan database" in exec mode

After creating all desired vlans use "exit" to apply the config

To view existing vlans use the command "show vlan-switch brief"

Alias(exec) : vl - "show vlan-switch brief" command

Alias(configure): va X - macro to add vlan X

Alias(configure): vd X - macro to delete vlan X

^C alias configure va macro global trace

add_vlan \$v alias configure vd macro global

trace del_vlan \$v alias exec vl show vlan-switch

brief

!

line con 0 exec-

timeout 0 0

privilege level 15

logging synchronous

line aux 0 exec-

timeout 0 0

privilege level 15

logging synchronous

line vty 0 4 login

!

!

end

VLAN Name Status Ports

1 default active Fa1/3, Fa1/4, Fa1/5, Fa1/6

Fa1/7, Fa1/8, Fa1/9, Fa1/10

Fa1/11, Fa1/12, Fa1/13, Fa1/14

Fa1/15

10 students active
20 profs active
1002 fddi-default act/unsup
1003 token-ring-default act/unsup
1004 fddinet-default act/unsup 1005 trnet-default act/unsup

Config SW4:

Building configuration... Current
configuration : 3176 bytes
!
version 12.4 service timestamps debug
datetime msec service timestamps log
datetime msec no service password-
encryption no service dhcp
!
hostname SW4
!
boot-start-marker boot-end-marker
!
!
no aaa new-model memory-size
iomem 5 no ip routing no ip
icmp rate-limit unreachable
no ip cef
!
!
!
!
no ip domain lookup
!
multilink bundle-name authenticated
!
!
!
!
!
!
!
!
!
!

!

!

!

!

!

!

!

!

!

macro name add_vlan

end vlan database

vlan \$v exit

@ macro name

del_vlan end vlan

database no vlan \$v

exit

@

!

vtp file nvram:vlan.dat

archive log config

hidekeys

!

!

!

!

ip tcp synwait-time 5

!

!

!

!

interface Port-channel2 switchport mode trunk

!

interface Port-channel1 switchport

mode trunk

!

interface FastEthernet0/0 description *** Unused
for Layer2 EtherSwitch ***

no ip address no

ip route-cache

shutdown

duplex auto

speed auto

```
!  
interface FastEthernet0/1 description *** Unused  
for Layer2 EtherSwitch ***  
no ip address no  
ip route-cache  
shutdown  
duplex auto  
speed auto  
!  
interface FastEthernet1/0  
switchport mode trunk  
duplex full speed 100  
channel-group 2 mode on  
!  
interface FastEthernet1/1  
switchport mode trunk  
duplex full speed 100  
channel-group 2 mode on  
!  
interface FastEthernet1/2  
switchport mode trunk  
duplex full speed 100  
channel-group 1 mode on  
!  
interface FastEthernet1/3  
switchport mode trunk  
duplex full speed 100  
channel-group 1 mode on  
interface FastEthernet1/4  
switchport mode trunk  
duplex full speed 100  
!  
interface FastEthernet1/5  
duplex full speed 100  
!  
interface FastEthernet1/6  
duplex full speed 100  
!  
interface FastEthernet1/7  
duplex full speed 100  
!
```

!

```
interface FastEthernet1/8
duplex full speed 100
!
interface FastEthernet1/9
duplex full speed 100
!
interface FastEthernet1/10
duplex full speed 100
!
interface FastEthernet1/11
duplex full speed 100
!
interface FastEthernet1/12
duplex full speed 100
!
interface FastEthernet1/13
duplex full speed 100
!
interface FastEthernet1/14 duplex
full
speed 100
!
interface FastEthernet1/15
duplex full speed
100
!
interface Vlan1
no ip address no
ip route-cache
shutdown
!
ip forward-protocol nd
!
!
no ip http server no ip
http secure-server
!
no cdp log mismatch duplex
!
!
!
!
```



```
!  
!  
control-plane
```

```
!  
!  
!  
!  
!  
!  
!  
!  
!
```

```
banner exec ^C
```

```
*****
```

This is a normal Router with a Switch module inside (NM-16ESW)
It has been pre-configured with hard-coded speed and duplex
To create vlans use the command "vlan database" in exec mode
After creating all desired vlans use "exit" to apply the config
To view existing vlans use the command "show vlan-switch brief"

Alias(exec) : vl - "show vlan-switch brief" command

Alias(configure): va X - macro to add vlan X

Alias(configure): vd X - macro to delete vlan X

^C alias configure va macro global trace

add_vlan \$v alias configure vd macro global

trace del_vlan \$v alias exec vl show vlan-switch

brief

!

line con 0 exec-

timeout 0 0

privilege level 15

logging synchronous

line aux 0 exec-

timeout 0 0

privilege level 15

logging synchronous

line vty 0 4 login

!

!

end

VLAN Name Status Ports

----- 1

default active Fa1/5, Fa1/6, Fa1/7, Fa1/8

Fa1/9, Fa1/10, Fa1/11, Fa1/12

Fa1/13, Fa1/14, Fa1/15

10 students active

20 profs active

1002 fddi-default act/unsup

1003 token-ring-default act/unsup

1004 fddinet-default act/unsup 1005 trnet-default act/unsup **Config SW5:**

Building configuration... Current

configuration : 2997 bytes

!

version 12.4

service timestamps debug datetime msec

service timestamps log datetime msec

no service password-encryption no

service dhcp

!

```
hostname SW5
!
boot-start-marker boot-end-marker
!
!
no aaa new-model memory-size
iomem 5 no ip routing no ip
icmp rate-limit unreachable
no ip cef
!
!
!
!
no ip domain lookup
!
multilink bundle-name authenticated
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
!
macro name add_vlan end
vlan database
vlan $v exit
@ macro name
del_vlan end vlan
```

```
database no vlan $v
exit
@
!
vtp file nvram:vlan.dat
archive log config
hidekeys
!
!
!
!
ip tcp synwait-time 5
!
!
!
!
interface FastEthernet0/0 description *** Unused
for Layer2 EtherSwitch ***
no ip address no
ip route-cache
shutdown
duplex auto
speed auto
!
interface FastEthernet0/1 description *** Unused
for Layer2 EtherSwitch ***
no ip address no
ip route-cache
shutdown
duplex auto
speed auto
!
interface FastEthernet1/0
switchport mode trunk
duplex full speed 100
```

!

```
interface FastEthernet1/1
duplex full speed 100
```

!

```
interface FastEthernet1/2
duplex full speed 100
```

!

```
interface FastEthernet1/3
duplex full speed 100
```

!

```
interface FastEthernet1/4
duplex full speed 100
```

!

```
interface FastEthernet1/5
duplex full speed 100
```

!

```
interface FastEthernet1/6
duplex full speed 100
```

!

```
interface FastEthernet1/7
duplex full speed 100
```

!

```
interface FastEthernet1/8
duplex full speed 100
```

!

```
interface FastEthernet1/9
duplex full speed 100
```

!

```
interface FastEthernet1/10
duplex full speed 100
```

!

```
interface FastEthernet1/11
duplex full speed 100
```

!

```
interface FastEthernet1/12
duplex full speed 100
```

!

```
interface FastEthernet1/13
switchport mode trunk
duplex full speed 100
```

!

```

interface FastEthernet1/14
switchport access vlan 20
duplex full speed 100
!
interface FastEthernet1/15
switchport access vlan 10
duplex full speed 100
!
interface Vlan1
no ip address no
ip route-cache
shutdown
!
ip default-gateway 192.168.99.254 ip
forward-protocol nd
!
!
no ip http server no ip
http secure-server
!
no cdp log mismatch duplex
!
!
!
!
!
!
control-plane
!
!
!
!
banner exec ^C
*****

```

This is a normal Router with a Switch module inside (NM-16ESW)
 It has been pre-configured with hard-coded speed and duplex
 To create vlans use the command "vlan database" in exec mode
 After creating all desired vlans use "exit" to apply the config
 To view existing vlans use the command "show vlan-switch brief"
 Alias(exec) : vl - "show vlan-switch brief" command
 Alias(configure): va X - macro to add vlan X

!
!
!
!
!

Alias(configure): vd X - macro to delete vlan X

^C alias configure va macro global trace
add_vlan \$v alias configure vd macro global
trace del_vlan \$v alias exec vl show vlan-switch
brief

!

line con 0 exec-
timeout 0 0
privilege level 15
logging synchronous
line aux 0 exec-
timeout 0 0
privilege level 15
logging synchronous
line vty 0 4 login

!

!

End

VLAN Name Status Ports

----- 1

default active Fa1/1, Fa1/2, Fa1/3, Fa1/4
Fa1/5, Fa1/6, Fa1/7, Fa1/8
Fa1/9, Fa1/10, Fa1/11, Fa1/12
10 students active Fa1/15
20 profs active Fa1/14
1002 fddi-default act/unsup
1003 token-ring-default act/unsup
1004 fddinet-default act/unsup 1005 trnet-default act/unsup

Config R1:

Building configuration... Current
configuration : 1692 bytes

!

version 12.4 service timestamps debug
datetime msec service timestamps log
datetime msec no service password-
encryption

!

hostname R1

```
!  
boot-start-marker boot-end-marker  
!  
!  
no aaa new-model memory-size  
iomem 5 no ip icmp rate-limit  
unreachable  
ip cef  
!  
!  
!  
!  
no ip domain lookup  
!  
multilink bundle-name authenticated  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
archive log  
config  
hidekeys  
!  
!  
!  
!  
ip tcp synwait-time 5  
!  
!
```



```
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
interface FastEthernet0/0 ip  
address 1.1.1.13 255.255.255.252  
duplex auto speed auto  
!  
interface FastEthernet0/1  
no ip address duplex  
auto speed auto  
!  
interface FastEthernet0/1.10 encapsulation  
dot1Q 10  
ip address 192.168.10.126 255.255.255.128  
!  
interface FastEthernet0/1.20 encapsulation  
dot1Q 20 ip address 192.168.20.126  
255.255.255.128  
!  
router bgp 200 no synchronization bgp log-  
neighbor-changes network 192.168.10.0  
mask 255.255.255.128 network  
192.168.20.0 mask 255.255.255.128  
neighbor 1.1.1.14 remote-as 100  
no auto-summary  
!  
ip forward-protocol nd  
!  
!  
no ip http server no ip  
http secure-server  
!  
no cdp log mismatch duplex  
!  
!
```

```
!  
!  
!  
!  
control-plane  
!  
!  
!  
!  
!  
line con 0 exec-  
timeout 0 0  
privilege level 15  
logging synchronous  
line aux 0 exec-  
timeout 0 0  
privilege level 15  
logging synchronous  
line vty 0 4 login  
!  
!  
end Config
```

R2:

```
Building configuration... Current  
configuration : 3253 bytes  
!  
version 12.4 service timestamps debug  
datetime msec service timestamps log  
datetime msec no service password-  
encryption  
!  
hostname R2  
!  
boot-start-marker boot-end-marker  
!  
!  
no aaa new-model memory-size  
iomem 5  
no ip icmp rate-limit unreachable  
ip cef
```

!
!
!
!
!
!
!

!

!

ip vrf vpn1 rd 100:1

route-target export 100:1

route-target import 100:1

!

no ip domain lookup

!

mpls traffic-eng tunnels multilink

bundle-name authenticated

!

!

!

!

!

!

!

!

!

!

!

!

!

!

!

!

!

!

!

!

!

archive log

config

hidekeys

!

!

!

!

ip tcp synwait-time 5

!

class-map match-any FTP-Traffic match

protocol ftp

!

!

```
policy-map FTP-QoS class
FTP-Traffic bandwidth
percent 50
```

!

!

!

!

!

```
interface Loopback0 ip address
2.2.2.2 255.255.255.255
```

!

```
interface Tunnel0 ip unnumbered Loopback0 tunnel
destination 3.3.3.3 tunnel mode mpls traffic-eng
tunnel mpls traffic-eng autoroute announce tunnel
mpls traffic-eng priority 1 1 tunnel mpls traffic-eng
bandwidth 200 tunnel mpls traffic-eng path-option 1
explicit name LSP1 no routing dynamic
```

!

```
interface Tunnel1 ip unnumbered Loopback0
tunnel destination 3.3.3.3 tunnel mode mpls
traffic-eng tunnel mpls traffic-eng autoroute
announce tunnel mpls traffic-eng priority 1 1
tunnel mpls traffic-eng bandwidth 200 tunnel
mpls traffic-eng path-option 1 dynamic no
routing dynamic
```

!

```
interface FastEthernet0/0 ip vrf
forwarding vpn1 ip address
1.1.1.14 255.255.255.252 duplex
auto speed auto service-policy
output FTP-QoS
```

!

```
interface FastEthernet0/1 ip
address 1.1.1.5 255.255.255.252
duplex auto speed auto mpls
label protocol ldp
mpls ip mpls traffic-eng
tunnels ip rsvp
bandwidth 300
```

!

```
interface Serial1/0 ip address
1.1.1.1 255.255.255.252 mpls
label protocol ldp mpls ip mpls
traffic-eng tunnels serial restart-
delay 0 ip rsvp bandwidth 300
!
interface Serial1/1
no ip address
shutdown
serial restart-delay 0
!
interface Serial1/2
no ip address
shutdown
serial restart-delay 0
!
interface Serial1/3
no ip address
shutdown
serial restart-delay 0
!
router ospf 1 mpls traffic-eng
router-id Loopback0 mpls traffic-
eng area 0 log-adjacency-changes
network 1.1.1.1 0.0.0.0 area 0
network 1.1.1.5 0.0.0.0 area 0
network 2.2.2.2 0.0.0.0 area 0
!
router bgp 100 no bgp default ipv4-
unicast bgp log-neighbor-changes
neighbor 3.3.3.3 remote-as 100 neighbor
3.3.3.3 update-source Loopback0
!
address-family vpnv4 neighbor 3.3.3.3
activate neighbor 3.3.3.3 send-community
extended
```

```
exit-address-family
!
address-family ipv4 vrf vpn1
neighbor 1.1.1.13 remote-as 200
neighbor 1.1.1.13 activate no
synchronization exit-address-
family
!
ip forward-protocol nd
!
!
no ip http server no ip
http secure-server
!
ip explicit-path name LSP1 enable
next-address 1.1.1.2 next-address
3.3.3.3
!
access-list 150 permit ip any any dscp af31
access-list 155 permit ip any any dscp af32 no
cdp log mismatch duplex
!
!
!
route-map vlan20-toTunnel permit 10
match ip address 155 set interface
Tunnel1
!
route-map vlan10-toTunnel permit 10
match ip address 150 set interface
Tunnel0
!
!
!
!
!
!
!
!
!
!
control-plane
!
!
!
!
!
!
!
!
!
```

```
!  
line con 0 exec-  
timeout 0 0  
privilege level 15  
logging synchronous  
line aux 0 exec-  
timeout 0 0  
privilege level 15  
logging synchronous  
line vty 0 4 login  
!  
!  
end
```

Config

R3:

Building configuration... Current
configuration : 3292 bytes

```
!  
version 12.4 service timestamps debug  
datetime msec service timestamps log  
datetime msec no service password-  
encryption  
!  
hostname R3  
!  
boot-start-marker boot-end-marker  
!  
!  
no aaa new-model memory-size  
iomem 5  
no ip icmp rate-limit unreachable  
ip cef  
!  
!  
!  
!  
ip vrf vpn1 rd 100:1  
route-target export 100:1  
route-target import  
100:1  
!  
no ip domain lookup  
!  
mpls traffic-eng tunnels multilink  
bundle-name authenticated
```



```
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
archive log  
config  
hidekeys  
!  
!  
!  
!  
ip tcp synwait-time 5  
!  
class-map match-any FTP-Traffic match  
protocol ftp  
!  
!  
policy-map FTP-QoS  
class FTP-Traffic bandwidth  
percent 50  
!  
!  
!  
!  
!  
interface Loopback0 ip address  
3.3.3.3 255.255.255.255
```

```
!  
interface Tunnel0 ip unnumbered Loopback0 tunnel  
destination 2.2.2.2 tunnel mode mpls traffic-eng  
tunnel mpls traffic-eng autoroute announce tunnel  
mpls traffic-eng priority 1 1 tunnel mpls traffic-eng  
bandwidth 200 tunnel mpls traffic-eng path-option 1  
explicit name LSP1 no routing dynamic
```

```
!  
interface Tunnel1 ip unnumbered Loopback0  
tunnel destination 2.2.2.2 tunnel mode mpls  
traffic-eng tunnel mpls traffic-eng autoroute  
announce tunnel mpls traffic-eng priority 1 1  
tunnel mpls traffic-eng bandwidth 200 tunnel  
mpls traffic-eng path-option 1 dynamic no  
routing dynamic
```

```
!  
interface FastEthernet0/0 ip vrf  
forwarding vpn1 ip address  
1.1.1.17 255.255.255.252 duplex  
auto speed auto
```

```
!  
interface FastEthernet0/1 ip  
address 1.1.1.9 255.255.255.252  
duplex auto speed auto mpls  
label protocol ldp mpls ip mpls  
traffic-eng tunnels service-policy  
output FTP-QoS ip rsvp bandwidth  
300
```

```
!  
interface Serial1/0  
ip address 1.1.1.2 255.255.255.252  
mpls label protocol ldp mpls ip  
mpls traffic-eng tunnels serial  
restart-delay 0 service-policy  
output FTP-QoS ip rsvp bandwidth  
300
```

```
!  
interface Serial1/1  
no ip address  
shutdown  
serial restart-delay 0
```

```
!  
interface Serial1/2 no  
ip address  
shutdown
```

```
serial restart-delay 0
!
interface Serial1/3
no ip address
shutdown
serial restart-delay 0
!
router ospf 1 mpls traffic-eng
router-id Loopback0 mpls traffic-
eng area 0 log-adjacency-changes
network 1.1.1.2 0.0.0.0 area 0
network 1.1.1.9 0.0.0.0 area 0
network 3.3.3.3 0.0.0.0 area 0
!
router bgp 100 no synchronization bgp
log-neighbor-changes neighbor 2.2.2.2
remote-as 100 neighbor 2.2.2.2 update-
source Loopback0
no auto-summary
!
address-family vpnv4 neighbor 2.2.2.2
activate neighbor 2.2.2.2 send-community
extended
exit-address-family
!
address-family ipv4 vrf vpn1
neighbor 1.1.1.18 remote-as 300
neighbor 1.1.1.18 activate no
synchronization exit-address-
family
!
ip forward-protocol nd
!
!
no ip http server no ip
http secure-server
!
ip explicit-path name LSP1 enable
next-address 1.1.1.1 next-address
2.2.2.2
!
access-list 150 permit ip any any dscp af31
access-list 155 permit ip any any dscp af32 no
cdp log mismatch duplex
!
```

```

!
!
route-map vlan20-toTunnel permit 10
match ip address 155 set interface
Tunnel1
!
route-map vlan10-toTunnel permit 10
match ip address 150 set interface
Tunnel0
!
!
!
!
control-plane
!
!
!
!
!
!
!
!
!
!
line con 0
exec-timeout 0 0
privilege level 15
logging synchronous
line aux 0 exec-
timeout 0 0
privilege level 15
logging synchronous
line vty 0 4 login
!
!
end

```

Config
R4:

Building configuration... Current
configuration : 1525 bytes

```

!
version 12.4 service timestamps debug
datetime msec service timestamps log
datetime msec no service password-
encryption

```

```
!  
hostname R4  
!  
boot-start-marker boot-end-marker  
!  
!  
no aaa new-model memory-size  
iomem 5  
no ip icmp rate-limit unreachable  
ip cef  
!  
!  
!  
!  
no ip domain lookup  
!  
mpls traffic-eng tunnels multilink  
bundle-name authenticated  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
archive log  
config  
hidekeys  
!  
!
```

```
!  
!  
ip tcp synwait-time 5  
!  
class-map match-any FTP-Traffic match  
protocol ftp  
!  
!  
policy-map FTP-QoS  
class FTP-Traffic bandwidth  
percent 50  
!  
!  
!  
!  
!  
interface Loopback0 ip address  
4.4.4.4 255.255.255.255  
!  
interface FastEthernet0/0  
ip address 1.1.1.10 255.255.255.252  
duplex auto speed auto mpls label  
protocol ldp mpls ip mpls traffic-  
eng tunnels ip rsvp bandwidth 300  
!  
interface FastEthernet0/1 ip  
address 1.1.1.6 255.255.255.252  
duplex auto speed auto mpls  
label protocol ldp mpls ip mpls  
traffic-eng tunnels service-policy  
output FTP-QoS ip rsvp bandwidth  
300  
!  
router ospf 1 mpls traffic-eng  
router-id Loopback0 mpls traffic-  
eng area 0 log-adjacency-changes  
network 1.1.1.6 0.0.0.0 area 0  
network 1.1.1.10 0.0.0.0 area 0  
network 4.4.4.4 0.0.0.0 area 0  
!  
ip forward-protocol nd  
!  
!  
no ip http server no ip  
http secure-server
```

!
no cdp log mismatch duplex

!

!

!

!

!

!

control-plane

!

!

!

!

!

!

!

!

!

!

line con 0 exec-timeout

0 0 privilege level 15

logging synchronous

line aux 0 exec-timeout

0 0 privilege level 15

logging synchronous

line vty 0 4 login

!

!

end **Config**

R5:

Building configuration... Current
configuration : 1696 bytes

!

version 12.4 service timestamps debug

datetime msec service timestamps log

datetime msec no service password-
encryption

!

hostname R5

!

boot-start-marker boot-end-marker

!

!

```
no aaa new-model memory-size  
iomem 5  
no ip icmp rate-limit unreachable  
ip cef  
!  
!  
!  
!  
no ip domain lookup  
!  
multilink bundle-name authenticated  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
archive log  
config  
hidekeys  
!  
!  
!  
!  
ip tcp synwait-time 5  
!  
class-map match-any FTP-Traffic  
match protocol ftp class-map  
match-any class20 match vlan
```



```
20 class-map match-any class10
match vlan 10
!
!
policy-map FTP-QoS class
FTP-Traffic bandwidth
percent 50 policy-map
vlan-tompls class class10
set dscp af31 class
class20 set dscp af32
!
!
!
!
!
interface FastEthernet0/0 ip
address 1.1.1.18 255.255.255.252
duplex auto speed auto service-
policy output FTP-QoS
!
interface FastEthernet0/1
no ip address duplex
auto speed auto
!
interface FastEthernet0/1.10 encapsulation
dot1Q 10 ip address 192.168.10.254
255.255.255.128 service-policy input vlan-
tompls
!
interface FastEthernet0/1.20 encapsulation
dot1Q 20 ip address 192.168.20.254
255.255.255.128 service-policy input vlan-
tompls
!
router bgp 300 no synchronization bgp log-
neighbor-changes network 192.168.10.128
mask 255.255.255.128 network
192.168.20.128 mask 255.255.255.128
neighbor 1.1.1.17 remote-as 100
no auto-summary
!
ip forward-protocol nd
!
!
no ip http server
```

```

no ip http secure-server
!
no cdp log mismatch duplex
!
!
!
!
!
!
control-plane
!
!
!
!
!
!
!
!
!
!
!
line con 0 exec-
timeout 0 0
privilege level 15
logging synchronous
line aux 0 exec-
timeout 0 0
privilege level 15
logging synchronous
line vty 0 4 login
!
!
end

```

2. Configuration de la deuxième partie du projet :

□Config SW1:

```

Building configuration... Current
configuration : 3041 bytes
!
version 12.4 service timestamps debug
datetime msec service timestamps log
datetime msec no service password-
encryption no service dhcp
!
hostname SW1

```

```
!  
boot-start-marker boot-end-marker  
!  
!  
no aaa new-model memory-size  
iomem 5 no ip routing no ip  
icmp rate-limit unreachable  
no ip cef  
!  
!  
!  
!  
no ip domain lookup  
!  
multilink bundle-name authenticated  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
macro name add_vlan end  
vlan database  
vlan $v exit  
@ macro name  
del_vlan end vlan  
database no vlan $v  
exit  
@  
!
```

```
vtp file nvram:vlan.dat
archive log config
hidekeys
!
!
!
!
ip tcp synwait-time 5
!
!
!
!
interface FastEthernet0/0 description *** Unused
for Layer2 EtherSwitch ***
no ip address no
ip route-cache
shutdown
duplex auto
speed auto
!
interface FastEthernet0/1 description *** Unused
for Layer2 EtherSwitch ***
no ip address no
ip route-cache
shutdown
duplex auto
speed auto
!
interface FastEthernet1/0 switchport trunk
allowed vlan 1,2,10,20,1002-1005 switchport
mode trunk duplex full
speed 100
!
interface FastEthernet1/1
switchport mode trunk
duplex full speed 100
!
interface FastEthernet1/2
duplex full speed 100
!
interface FastEthernet1/3
duplex full speed 100
!
interface FastEthernet1/4
duplex full speed 100
```

```
!  
interface FastEthernet1/5  
duplex full speed 100  
!  
interface FastEthernet1/6  
duplex full speed 100  
!  
interface FastEthernet1/7  
duplex full speed 100  
!  
interface FastEthernet1/8  
duplex full speed 100  
!  
interface FastEthernet1/9  
duplex full speed 100  
!  
interface FastEthernet1/10  
duplex full speed 100  
!  
interface FastEthernet1/11 duplex full speed 100  
!  
interface FastEthernet1/12  
duplex full speed 100  
!  
interface FastEthernet1/13  
switchport access vlan 99  
duplex full speed 100  
!  
interface FastEthernet1/14  
switchport access vlan 10  
duplex full speed 100  
!  
interface FastEthernet1/15  
switchport access vlan 20  
duplex full speed 100  
!  
interface Vlan1  
no ip address no  
ip route-cache  
shutdown  
!  
ip forward-protocol nd  
!  
!  
no ip http server no ip  
http secure-server
```

```

!
no cdp log mismatch duplex
!
!
!
!
!
!
control-plane
!
!
!
!
!
!
!
!
!
!
banner exec ^C
*****

This is a normal Router with a Switch module inside (NM-16ESW)
It has been pre-configured with hard-coded speed and duplex
To create vlans use the command "vlan database" in exec mode
After creating all desired vlans use "exit" to apply the config
To view existing vlans use the command "show vlan-switch brief"
Alias(exec) : vl - "show vlan-switch brief" command
Alias(configure): va X - macro to add vlan X
Alias(configure): vd X - macro to delete vlan X
*****

^C alias configure va macro global trace
add_vlan $v alias configure vd macro global
trace del_vlan $v alias exec vl show vlan-switch
brief
!
line con 0 exec-
timeout 0 0
privilege level 15
logging synchronous
line aux 0 exec-
timeout 0 0
privilege level 15
logging synchronous
line vty 0 4 login
!
!

```

end

VLAN Name Status Ports

----- 1

default active Fa1/2, Fa1/3, Fa1/4, Fa1/5

Fa1/6, Fa1/7, Fa1/8, Fa1/9

Fa1/10, Fa1/11, Fa1/12

10 students active Fa1/14

20 profs active Fa1/15

1002 fddi-default act/unsup

1003 token-ring-default act/unsup

1004 fddinet-default act/unsup 1005 trnet-default act/unsup **Config SW2:**

Building configuration... Current

configuration : 3008 bytes

!

version 12.4 service timestamps debug

datetime msec service timestamps log

datetime msec no service password-

encryption no service dhcp

!

hostname SW2

!

boot-start-marker boot-end-marker

!

!

no aaa new-model memory-size

iomem 5 no ip routing no ip

icmp rate-limit unreachable

no ip cef

!

!

!

!

no ip domain lookup

!

multilink bundle-name authenticated

!

!

!

!

!

!

!

!

```
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
macro name add_vlan  
end vlan database  
vlan $v exit  
@ macro name  
del_vlan end vlan  
database no vlan $v  
exit  
@  
!  
vtp file nvram:vlan.dat  
archive log config  
hidekeys  
!  
!  
!  
!  
ip tcp synwait-time 5  
!  
!  
!  
!  
interface Port-channel1 switchport  
mode trunk  
!  
interface FastEthernet0/0 description *** Unused for Layer2 EtherSwitch  
***  
no ip address no  
ip route-cache  
shutdown  
duplex auto  
speed auto  
!
```



```
interface FastEthernet0/1 description *** Unused
for Layer2 EtherSwitch ***
no ip address no
ip route-cache
shutdown
duplex auto
speed auto
!
interface FastEthernet1/0
switchport mode trunk
duplex full speed 100
channel-group 1 mode on
!
interface FastEthernet1/1
switchport mode trunk
duplex full speed 100
channel-group 1 mode on
!
interface FastEthernet1/2
duplex full speed 100
!
interface FastEthernet1/3
duplex full speed 100
!
interface FastEthernet1/4
duplex full speed 100
!
interface FastEthernet1/5
duplex full speed 100
!
interface FastEthernet1/6
duplex full speed
100
!
interface FastEthernet1/7
duplex full speed 100
!
interface FastEthernet1/8
duplex full speed 100
!
interface FastEthernet1/9
duplex full speed 100
!
interface FastEthernet1/10
duplex full speed 100
!
```

```
interface FastEthernet1/11
duplex full speed 100
!
interface FastEthernet1/12
duplex full speed 100
!
interface FastEthernet1/13
duplex full speed 100
!
interface FastEthernet1/14
duplex full speed 100
!
interface FastEthernet1/15
duplex full speed 100
!
interface Vlan1
no ip address no
ip route-cache
shutdown
!
ip forward-protocol nd
!
!
no ip http server no ip
http secure-server
!
no cdp log mismatch duplex
!
!
!
!
!
!
!
control-plane
!
!
!
!
!
!
!
!
!
banner exec ^C
```

This is a normal Router with a Switch module inside (NM-16ESW)

It has been pre-configured with hard-coded speed and duplex

To create vlans use the command "vlan database" in exec mode

After creating all desired vlans use "exit" to apply the config

To view existing vlans use the command "show vlan-switch brief"

Alias(exec) : vl - "show vlan-switch brief" command

Alias(configure): va X - macro to add vlan X

Alias(configure): vd X - macro to delete vlan X

^C

alias configure va macro global trace add_vlan \$v

alias configure vd macro global trace del_vlan \$v alias

exec vl show vlan-switch brief

!

line con 0 exec-

timeout 0 0

privilege level 15

logging synchronous

line aux 0 exec-

timeout 0 0

privilege level 15

logging synchronous

line vty 0 4 login

!

!

end

VLAN Name Status Ports

----- 1

default active Fa1/2, Fa1/3, Fa1/4, Fa1/5

Fa1/6, Fa1/7, Fa1/8, Fa1/9

Fa1/10, Fa1/11, Fa1/12, Fa1/13

Fa1/14, Fa1/15

10 students active

20 profs active

1002 fddi-default act/unsup

1003 token-ring-default act/unsup

1004 fddinet-default act/unsup 1005 trnet-default act/unsup

Config SW3:

Building configuration... Current

configuration : 3031 bytes

!

version 12.4 service timestamps debug

datetime msec service timestamps log

datetime msec no service password-

encryption no service dhcp

```
!  
hostname SW3  
!  
boot-start-marker boot-end-marker  
!  
!  
no aaa new-model memory-size  
iomem 5 no ip routing no ip  
icmp rate-limit unreachable  
no ip cef  
!  
!  
!  
!  
no ip domain lookup  
!  
multilink bundle-name authenticated  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
macro name add_vlan  
end vlan database  
vlan $v  
exit  
@ macro name  
del_vlan end vlan
```

```
database no vlan $v
exit
@
!
vtp file nvram:vlan.dat
archive log config
hidekeys
!
!
!
!
ip tcp synwait-time 5
!
!
!
!
interface Port-channel2 switchport
mode trunk
!
interface FastEthernet0/0 description *** Unused
for Layer2 EtherSwitch ***
no ip address no
ip route-cache
shutdown
duplex auto
speed auto
!
interface FastEthernet0/1 description *** Unused
for Layer2 EtherSwitch ***
no ip address no
ip route-cache
shutdown
duplex auto
speed auto
!
interface FastEthernet1/0
switchport mode trunk
duplex full speed 100
!
interface FastEthernet1/1
switchport mode trunk
duplex full speed 100
channel-group 2 mode on
!
```

```
interface FastEthernet1/2
switchport mode trunk
duplex full speed 100
channel-group 2 mode on
!
interface FastEthernet1/3
duplex full speed 100
!
interface FastEthernet1/4
duplex full speed 100
!
interface FastEthernet1/5
duplex full speed 100
!
interface FastEthernet1/6
duplex full speed 100
!
interface FastEthernet1/7
duplex full speed 100
!
interface FastEthernet1/8
duplex full speed 100
!
interface FastEthernet1/9
duplex full speed 100
!
interface FastEthernet1/10
duplex full speed 100
!
interface FastEthernet1/11
duplex full speed 100
!
interface FastEthernet1/12 duplex
full
speed 100
!
interface FastEthernet1/13
duplex full speed 100
!
interface FastEthernet1/14
duplex full speed 100
!
interface FastEthernet1/15
duplex full speed 100
!
```

```

interface Vlan1
no ip address no
ip route-cache
shutdown
!
ip forward-protocol nd
!
!
no ip http server no ip
http secure-server
!
no cdp log mismatch duplex
!
!
!
!
!
!
control-plane
!
!
!
!
!
!
!
!
!
banner exec ^C
*****

This is a normal Router with a Switch module inside (NM-16ESW)
It has been pre-configured with hard-coded speed and duplex
To create vlans use the command "vlan database" in exec mode
After creating all desired vlans use "exit" to apply the config
To view existing vlans use the command "show vlan-switch brief"
Alias(exec) : vl - "show vlan-switch brief" command
Alias(configure): va X - macro to add vlan X
Alias(configure): vd X - macro to delete vlan X
*****

^C alias configure va macro global trace
add_vlan $v alias configure vd macro global
trace del_vlan $v alias exec vl show vlan-switch
brief
!

```

```
line con 0 exec-
timeout 0 0
privilege level 15
logging synchronous
line aux 0 exec-
timeout 0 0
privilege level 15
logging synchronous
line vty 0 4 login
```

```
!
```

```
!
```

```
end
```

```
VLAN Name Status Ports
```

```
----- 1
```

```
default active Fa1/3, Fa1/4, Fa1/5, Fa1/6
Fa1/7, Fa1/8, Fa1/9, Fa1/10
Fa1/11, Fa1/12, Fa1/13, Fa1/14
Fa1/15
```

```
10 students active
```

```
20 profs active
```

```
1002 fddi-default act/unsup
```

```
1003 token-ring-default act/unsup
```

```
1004 fddinet-default act/unsup
```

```
1005 trnet-default act/unsup
```

❏ Config SW4:

```
Building configuration... Current
configuration : 3176 bytes
```

```
!
```

```
version 12.4 service timestamps debug
```

```
datetime msec service timestamps log
```

```
datetime msec no service password-
```

```
encryption no service dhcp
```

```
!
```

```
hostname SW4
```

```
!
```

```
boot-start-marker boot-end-marker
```

```
!
```

```
!
```

```
no aaa new-model memory-size
```

```
iomem 5 no ip routing no ip
```

```
icmp rate-limit unreachable
```

```
no ip cef
```

```
!
```



```
!  
!  
!  
no ip domain lookup  
!  
multilink bundle-name authenticated  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
macro name add_vlan  
end vlan database  
vlan $v exit  
@ macro name  
del_vlan end vlan  
database no vlan $v  
exit  
@  
!  
vtp file nvram:vlan.dat  
archive log config  
hidekeys  
!  
!  
!  
!  
ip tcp synwait-time 5  
!
```

```
!  
!  
!  
interface Port-channel2 switchport  
mode trunk  
!  
interface Port-channel1 switchport  
mode trunk  
!  
interface FastEthernet0/0 description *** Unused  
for Layer2 EtherSwitch ***  
no ip address no  
ip route-cache  
shutdown  
duplex auto speed  
auto  
!  
interface FastEthernet0/1 description *** Unused  
for Layer2 EtherSwitch ***  
no ip address no  
ip route-cache  
shutdown  
duplex auto  
speed auto  
!  
interface FastEthernet1/0  
switchport mode trunk  
duplex full speed 100  
channel-group 2 mode on  
!  
interface FastEthernet1/1  
switchport mode trunk  
duplex full speed 100  
channel-group 2 mode on  
!  
interface FastEthernet1/2  
switchport mode trunk  
duplex full speed 100  
channel-group 1 mode on  
!  
interface FastEthernet1/3  
switchport mode trunk  
duplex full speed 100  
channel-group 1 mode on  
!
```

```
interface FastEthernet1/4
switchport mode trunk
duplex full speed 100
!
interface FastEthernet1/5
duplex full speed 100
!
interface FastEthernet1/6
duplex full speed 100
!
interface FastEthernet1/7
duplex full speed 100
!
interface FastEthernet1/8
duplex full speed 100
!
interface FastEthernet1/9
duplex full speed 100
!
interface FastEthernet1/10
duplex full speed 100
!
interface FastEthernet1/11
duplex full speed 100
!
interface FastEthernet1/12
duplex full speed 100
!
interface FastEthernet1/13
duplex full speed 100
!
interface FastEthernet1/14
duplex full speed 100
!
interface FastEthernet1/15
duplex full speed 100
!
interface Vlan1
no ip address no
ip route-cache
shutdown
!
ip forward-protocol nd
!
!
```

```

no ip http server no ip
http secure-server
!
no cdp log mismatch duplex
!
!
!
!
!
!
control-plane
!
!
!
!
!
!
!
!
!
!
banner exec ^C
*****

This is a normal Router with a Switch module inside (NM-16ESW)
It has been pre-configured with hard-coded speed and duplex
To create vlans use the command "vlan database" in exec mode
After creating all desired vlans use "exit" to apply the config
To view existing vlans use the command "show vlan-switch brief"
Alias(exec) : vl - "show vlan-switch brief" command
Alias(configure): va X - macro to add vlan X
Alias(configure): vd X - macro to delete vlan X
*****

^C
alias configure va macro global trace add_vlan $v
alias configure vd macro global trace del_vlan $v alias
exec vl show vlan-switch brief
!
line con 0 exec-
timeout 0 0
privilege level 15
logging synchronous
line aux 0 exec-
timeout 0 0
privilege level 15
logging synchronous
line vty 0 4 login

```

```

!
!
end
VLAN Name Status Ports
-----
default active Fa1/5, Fa1/6, Fa1/7, Fa1/8
Fa1/9, Fa1/10, Fa1/11, Fa1/12
Fa1/13, Fa1/14, Fa1/15
10 students active
20 profs active
1002 fddi-default act/unsup
1003 token-ring-default act/unsup
1004 fddinet-default act/unsup 1005 trnet-default act/unsup

```

Config SW5:

Building configuration... Current
configuration : 2997 bytes

```

!
version 12.4 service timestamps debug
datetime msec service timestamps log
datetime msec no service password-
encryption no service dhcp
!
hostname SW5
!
boot-start-marker boot-end-marker
!
!
no aaa new-model memory-size
iomem 5 no ip routing no ip
icmp rate-limit unreachable
no ip cef
!
!
!
!
no ip domain lookup
!
multilink bundle-name authenticated
!
!
!
!
!
!

```

```
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
macro name add_vlan  
end vlan database  
vlan $v  
exit  
@ macro name  
del_vlan end vlan  
database no vlan $v  
exit  
@  
!  
vtp file nvram:vlan.dat  
archive log config  
hidekeys  
!  
!  
!  
!  
ip tcp synwait-time 5  
!  
!  
!  
!  
interface FastEthernet0/0 description *** Unused  
for Layer2 EtherSwitch ***  
no ip address no  
ip route-cache  
shutdown  
duplex auto  
speed auto  
!
```

```
interface FastEthernet0/1 description *** Unused
for Layer2 EtherSwitch ***
no ip address no
ip route-cache
shutdown
duplex auto
speed auto
!
interface FastEthernet1/0
switchport mode trunk
duplex full speed 100
!
interface FastEthernet1/1
duplex full speed 100
!
interface FastEthernet1/2
duplex full speed 100
!
interface FastEthernet1/3
duplex full speed
100
!
interface FastEthernet1/4
duplex full speed 100
!
interface FastEthernet1/5
duplex full speed 100
!
interface FastEthernet1/6
duplex full speed 100
!
interface FastEthernet1/7
duplex full speed 100
!
interface FastEthernet1/8
duplex full speed 100
!
interface FastEthernet1/9
duplex full speed 100
!
interface FastEthernet1/10
duplex full speed 100
!
interface FastEthernet1/11
duplex full speed 100
!
```

```
interface FastEthernet1/12
duplex full speed 100
!
interface FastEthernet1/13
switchport mode trunk
duplex full speed 100
!
interface FastEthernet1/14 switchport
access vlan 20
duplex full speed
100
!
interface FastEthernet1/15 switchport
access vlan 10
duplex full speed
100
!
interface Vlan1
no ip address no
ip route-cache
shutdown
!
ip default-gateway 192.168.99.254 ip
forward-protocol nd
!
!
no ip http server no ip
http secure-server
!
no cdp log mismatch duplex
!
!
!
!
!
!
control-plane
!
!
!
!
!
!
!
```



```

!
banner exec ^C
*****

This is a normal Router with a Switch module inside (NM-16ESW)
It has been pre-configured with hard-coded speed and duplex
To create vlans use the command "vlan database" in exec mode
After creating all desired vlans use "exit" to apply the config
To view existing vlans use the command "show vlan-switch brief"
Alias(exec) : vl - "show vlan-switch brief" command
Alias(configure): va X - macro to add vlan X
Alias(configure): vd X - macro to delete vlan X
*****

^C alias configure va macro global trace
add_vlan $v alias configure vd macro global
trace del_vlan $v alias exec vl show vlan-switch
brief
!
line con 0 exec-
timeout 0 0
privilege level 15
logging synchronous
line aux 0 exec-
timeout 0 0
privilege level 15
logging synchronous
line vty 0 4 login
!
!
End
VLAN Name Status Ports
-----
default active Fa1/1, Fa1/2, Fa1/3, Fa1/4
Fa1/5, Fa1/6, Fa1/7, Fa1/8
Fa1/9, Fa1/10, Fa1/11, Fa1/12
10 students active Fa1/15
20 profs active Fa1/14
1002 fddi-default act/unsup
1003 token-ring-default act/unsup
1004 fddinet-default act/unsup 1005 trnet-default act/unsup
Config R1:

Building configuration...
Current configuration : 1434 bytes
!
version 12.4 service timestamps debug
datetime msec service timestamps log

```

```
datetime msec no service password-
encryption
```

!

```
hostname R1
```

!

boot-start-marker boot-end-marker

!

!

```
no aaa new-model memory-size
```

iomem 5

```
no ip icmp rate-limit unreachable
```

ip cef

!

!

!

!

no ip domain lookup

!

multilink bundle-name authenticated

!

!

!

!

!

!

!

!

!

!

!

!

!

!

!

!

!

1

1

1

1

archive log

config

hidekeys

!

```
!  
!  
!  
ip tcp synwait-time 5  
pseudowire-class L2vpn  
encapsulation mpls interworking  
ethernet  
!  
!  
!  
!  
!  
interface Loopback0 ip address  
11.11.11.11 255.255.255.255 ip ospf  
network point-to-point  
!  
interface FastEthernet0/0 ip  
address 1.1.1.1 255.255.255.252  
duplex auto speed auto mpls  
label protocol ldp mpls ip  
!  
interface FastEthernet0/1  
no ip address duplex  
auto speed auto  
!  
interface FastEthernet0/1.10 encapsulation dot1Q 10  
xconnect 55.55.55.55 10 encapsulation mpls pw-class L2vpn  
!  
interface FastEthernet0/1.20 encapsulation dot1Q 20  
xconnect 55.55.55.55 20 encapsulation mpls pw-class L2vpn  
!  
router ospf 1 log-adjacency-changes  
network 1.1.1.1 0.0.0.0 area 0  
network 11.11.11.11 0.0.0.0 area 0  
!  
ip forward-protocol nd  
!  
!  
no ip http server no ip  
http secure-server  
!  
no cdp log mismatch duplex  
!  
!  
!
```

```
!  
!  
!  
control-plane  
!  
!  
!  
!  
!  
!  
!  
!  
!  
line con 0 exec-  
timeout 0 0  
privilege level 15  
logging synchronous  
line aux 0 exec-  
timeout 0 0  
privilege level 15  
logging synchronous  
line vty 0 4 login  
!  
!  
end Config  
R2:
```

```
Building configuration...  
Current configuration : 1554 bytes  
!  
version 12.4 service timestamps debug  
datetime msec service timestamps log  
datetime msec no service password-  
encryption  
!  
hostname R2  
!  
boot-start-marker boot-end-marker  
!  
!  
no aaa new-model memory-size  
iomem 5  
no ip icmp rate-limit unreachable  
ip cef
```

```
!  
!  
!  
!  
no ip domain lookup  
!  
multilink bundle-name authenticated  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
archive log  
config  
hidekeys  
!  
!  
!  
!  
ip tcp synwait-time 5  
!  
!  
!  
!  
interface Loopback0 ip address  
22.22.22.22 255.255.255.255  
!
```

```
interface FastEthernet0/0 ip
address 1.1.1.2 255.255.255.252
duplex auto speed auto mpls
label protocol ldp mpls ip
!
interface FastEthernet0/1 ip
address 1.1.1.5 255.255.255.252
duplex auto speed auto mpls
label protocol ldp mpls ip
!
interface Serial1/0 ip address
1.1.1.9 255.255.255.252 mpls
label protocol ldp mpls ip serial
restart-delay 0
!
interface Serial1/1 no
ip address
shutdown
serial restart-delay 0
!
interface Serial1/2 no
ip address shutdown
serial restart-delay 0
!
interface Serial1/3
no ip address
shutdown
serial restart-delay 0
!
router ospf 1 log-adjacency-
changes network 1.1.1.2 0.0.0.0
area 0 network 1.1.1.5 0.0.0.0
area 0 network 1.1.1.9 0.0.0.0
area 0 network 22.22.22.22
0.0.0.0 area 0
!
ip forward-protocol nd
!
!
no ip http server no ip
http secure-server
!
no cdp log mismatch duplex
!
!
```

```
!  
!  
!  
!  
control-plane  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
line con 0 exec-timeout  
0 0 privilege level 15  
logging synchronous  
line aux 0 exec-timeout  
0 0 privilege level 15  
logging synchronous  
line vty 0 4 login  
!  
!  
end
```

Config

R3:

Building configuration... Current
configuration : 1560 bytes

```
!  
version 12.4 service timestamps debug  
datetime msec service timestamps log  
datetime msec no service password-  
encryption  
!  
hostname R3  
!  
boot-start-marker boot-end-marker  
!  
!  
no aaa new-model memory-size  
iomem 5  
no ip icmp rate-limit unreachable  
ip cef  
!
```

```
!  
!  
!  
no ip domain lookup  
!  
multilink bundle-name authenticated  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
archive log  
config  
hidekeys  
!  
!  
!  
!  
ip tcp synwait-time 5  
!  
!  
!  
!  
interface Loopback0 ip address  
33.33.33.33 255.255.255.255  
!  
interface FastEthernet0/0 ip  
address 1.1.1.14 255.255.255.252
```



```
duplex auto speed auto mpls label
protocol ldp mpls ip
!
interface FastEthernet0/1 ip
address 1.1.1.17 255.255.255.252
duplex auto speed auto mpls label
protocol ldp mpls ip
!
interface Serial1/0
ip address 1.1.1.10 255.255.255.252
mpls label protocol ldp mpls ip
serial restart-delay 0
!
interface Serial1/1
no ip address
shutdown
serial restart-delay 0
!
interface Serial1/2
no ip address
shutdown
serial restart-delay 0
!
interface Serial1/3
no ip address
shutdown
serial restart-delay 0
!
router ospf 1 log-adjacency-
changes network 1.1.1.10 0.0.0.0
area 0 network 1.1.1.14 0.0.0.0
area 0 network 1.1.1.17 0.0.0.0
area 0 network 33.33.33.33
0.0.0.0 area 0
!
ip forward-protocol nd
!
!
no ip http server no ip
http secure-server
!
no cdp log mismatch duplex
!
!
!
```

```
!  
!  
!  
control-plane  
!  
!  
!  
!  
!  
!  
!  
!  
!  
line con 0 exec-  
timeout 0 0  
privilege level 15  
logging synchronous  
line aux 0 exec-  
timeout 0 0  
privilege level 15  
logging synchronous  
line vty 0 4 login  
!  
!  
end Config  
R4:
```

Building configuration... Current
configuration : 1195 bytes

```
!  
version 12.4 service timestamps debug  
datetime msec service timestamps log  
datetime msec no service password-  
encryption  
!  
hostname R4  
!  
boot-start-marker boot-end-marker  
!  
!  
no aaa new-model memory-size  
iomem 5  
no ip icmp rate-limit unreachable  
ip cef
```

```
!  
!  
!  
!  
no ip domain lookup  
!  
multilink bundle-name authenticated  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
archive log  
config  
hidekeys  
!  
!  
!  
!  
ip tcp synwait-time 5  
!  
!  
!  
!  
interface Loopback0 ip address  
44.44.44.44 255.255.255.255  
!
```

```
interface FastEthernet0/0 ip
address 1.1.1.6 255.255.255.252
duplex auto
speed auto mpls label
protocol ldp mpls ip
!
interface FastEthernet0/1 ip
address 1.1.1.13 255.255.255.252
duplex auto speed auto mpls label
protocol ldp mpls ip
!
router ospf 1 log-adjacency-
changes network 1.1.1.6 0.0.0.0
area 0 network 1.1.1.13 0.0.0.0
area 0 network 44.44.44.44
0.0.0.0 area 0
!
ip forward-protocol nd
!
!
no ip http server no ip
http secure-server
!
no cdp log mismatch duplex
!
!
!
!
!
!
!
control-plane
!
!
!
!
!
!
!
!
!
!
line con 0 exec-timeout
0 0 privilege level 15
logging synchronous
line aux 0 exec-timeout
```

0 0 privilege level 15
logging synchronous
line vty 0 4 login

!

!

end **Config**

R5:

Building configuration... Current
configuration : 1436 bytes

!

version 12.4 service timestamps debug
datetime msec service timestamps log
datetime msec no service password-
encryption

!

hostname R5

!

boot-start-marker boot-end-marker

!

!

no aaa new-model memory-size
iomem 5

no ip icmp rate-limit unreachable

ip cef

!

!

!

no ip domain lookup

!

multilink bundle-name authenticated

!

!

!

!

!

!

!

archive log
config

hidekeys

!

!

```
!  
!  
ip tcp synwait-time 5  
pseudowire-class L2vpn  
encapsulation mpls interworking  
ethernet  
!  
!  
!  
!  
!  
interface Loopback0 ip address  
55.55.55.55 255.255.255.255 ip ospf  
network point-to-point  
!  
interface FastEthernet0/0 ip  
address 1.1.1.18 255.255.255.252  
duplex auto speed auto mpls label  
protocol ldp mpls ip  
!  
interface FastEthernet0/1  
no ip address duplex  
auto speed auto  
!  
!  
router ospf 1 log-adjacency-  
changes network 1.1.1.18 0.0.0.0  
area 0 network 55.55.55.55  
0.0.0.0 area 0  
!  
ip forward-protocol nd  
!  
!  
no ip http server no ip  
http secure-server  
!  
no cdp log mismatch duplex  
!  
!  
!  
!  
!  
control-plane  
!!
```

```
!  
!  
!  
!  
!  
!  
!  
line con 0 exec-timeout  
0 0 privilege level 15  
logging synchronous  
line aux 0 exec-timeout  
0 0 privilege level 15  
logging synchronous  
line vty 0 4 login  
!  
!  
end
```